

EXERCISE

CONSIDER THE SSP GENERATED BY

$$S: y(t) = e(t) + \frac{1}{2} e(t-1) \quad e \sim WN(0, 1)$$

AND THE MODEL CLASS

$$M: y(t) = \eta(t) + a \eta(t-1) \quad \eta \sim WN(0, \lambda^2) \quad MA(1)$$

⇒ COMPUTE THE VALUE θ^* OF THE VECTOR OF PARAMETERS θ WHICH MINIMIZES THE COST FUNCTION

$$J(\theta) = E[(y(t) - \hat{y}(t|t-2))^2]$$

NOTE: THIS IS NOT THE P.G.M. CRITERION!

↳ 2-STEP AHEAD PREDICTION

1. Predictor

$$W(z) = \frac{1 + az^{-1}}{1} = E(z) + \frac{R(z)}{A(z)} \rightarrow \begin{cases} E(z) = 1 + az^{-1} \\ R(z) = 0 \end{cases}$$

THUS THE PREDICTOR IS:

$$\hat{y}(t|t-2) = \frac{R(z)}{C(z)} \quad y(t) = 0$$

NOTE: SINCE THE PROCESS IS MA(1)

IT "LOSES MEMORY" AFTER 2 TIME STEPS.

THE PREDICTOR IS TRIVIAL

2. COST FUNCTION DEFINITION

$$J(\theta) = E[y(t)^2] = E[(e(t) + \frac{1}{2} e(t-1))^2]$$



IT IS NOT FUNCTION OF θ
SO ANY VALUE OF "a" WOULD BE OK!



THE P.G.M. METHOD RELIES ON 1-STEP AHEAD PREDICTION

EXERCISE - VARIANT OF THE PREVIOUS

CONSIDER A DIFFERENT MODEL CLASS

$$M: y(t) = y(t) + a y(t-2) \quad y \sim WN(0, \lambda^2) \quad MA(1)$$

IDENTIFY THE PARAMETER $\theta = a$ MINIMIZING 1-STEP AHEAD PREDICTION ERROR

1. PREDICTOR

$$W(z) = 1 + a z^{-2} \rightarrow \hat{y}(t|t-1) = \frac{a z^{-2}}{1 + a z^{-2}} y(t)$$

PREDICTOR

↓ RECURSIVE EQUATION

$$\hat{y}(t|t-1) = -a \hat{y}(t-2|t-3) + a y(t-2)$$

NOTE: CONSIDERING THE SPECIFIED MODEL CLASS, IT'S REASONABLE THAT THERE IS NO INFLUENCE OF DATA $y(t-1)$ AS THEY DO NOT AFFECT $y(t)$.

2. COST FUNCTION

THE P.B.M. COST FUNCTION IS THE PREDICTION ERROR $\varepsilon(t|t-1)$ VARIANCE. FIRST, WE FIND THE STOCHASTIC DESCRIPTION OF THE PREDICTION ERROR!

$$\varepsilon(t) = \left[1 - \frac{a z^{-2}}{1 + a z^{-2}} \right] y(t) = \frac{1}{1 + a z^{-2}} y(t)$$

$\underbrace{\hspace{1cm}}_{y(t+1)} \quad \underbrace{\hspace{1cm}}_{\hat{y}(t+1|t)}$

$\hookrightarrow \varepsilon(t) = -a \varepsilon(t-2) + y(t)$

$$\varepsilon(t) = -a \varepsilon(t-2) + e(t) + \frac{1}{2} e(t-1) \quad \text{ARMA}$$

THUS, THE COST FUNCTION IS:

$$J(\theta) = \mathbb{E}[(y(t) - \hat{y}(t|t-1))^2] = \mathbb{E}[\varepsilon^2(t)] = \text{Var}[\varepsilon(t)]$$

DEFINITION OF $\varepsilon(t)$ AS STOCHASTIC PROCESS

$$= a^2 \mathbb{E}[\varepsilon^2(t-2)] + \mathbb{E}[e^2(t)] + \frac{1}{4} \mathbb{E}[e^2(t-1)] +$$

$$- 2a \mathbb{E}[\varepsilon(t-2) e(t)] - a \mathbb{E}[\varepsilon(t-2) e(t-1)] + \mathbb{E}[e(t) e(t-1)]$$

$\hookrightarrow f(e(t-2), \dots)$ $\hookrightarrow f(e(t-2), \dots)$

$$\gamma_\varepsilon(0) = a^2 \gamma_\varepsilon(0) + 1 + \frac{1}{4}$$

$$\Rightarrow J(\theta) = \frac{5/4}{1 - a^2}$$

3. Minimize $J(\theta)$

$$\frac{\partial J}{\partial \theta} = \frac{+5/4 \cdot 2a}{(1-a^2)^2} = \frac{5/2 a}{(1-a^2)^2} \Rightarrow \frac{\partial J}{\partial \theta} = 0, \quad a = 0$$

Notes: $\text{Var}[E(y)] = J(\theta^*) = 5/4 > \lambda^2 = 1 \Rightarrow$ This is consistent with the fact that we don't use the correct noise class